1.3 Chemical Bonding

Question Paper

Course	CIE A Level Chemistry
Section	1. Physical Chemistry
Topic	1.3 Chemical Bonding
Difficulty	Hard

Time allowed: 70

Score: /57

Percentage: /100

Question 1a

This question is about various polyatomic ions.

The nitrate(V) ion, NO_3^-, is a polyatomic ion, bonded by covalent bonds.
The three oxygen atoms are bonded by one single covalent bond, one double covalent bond and one coordinate bond.

Draw the dot-and-cross diagram for NO_3^-.

[2 marks]

Question 1b

An ionic compound has the empirical formula $H_4N_2O_3$.

Suggest the formulae of the ions present in this compound.

[2 marks]

Question 1c

Cyanide is a fast-acting chemical, which can be found in various forms and can have toxic effects on the body.

Draw the dot-and-cross diagram for a CN^- ion.

Show the outer electrons only.

[2 marks]

Question 1d

i)

Compare the average bond enthalpy in the cyanide ion to the average bond enthalpy of the C-N bond in methylamine, CH_3NH_2. Explain your answer.

[2]

ii)

Explain why the C-N bond length in the cyanide ion is shorter than in methylamine.

[2]

[4 marks]

Question 2a

Amides are commonly used in industrial processes as solvents, and there are suggestions that the smallest amide, methanamide, $HCONH_2$, could function in a similar manner to water as a solvent for life on other planets.

Draw the dot-and-cross diagram for methanamide, $HCONH_2$.

[2 marks]

Question 2b

Predict and explain the H-C-N and C-N-H bond angles around the C and N atoms.

[6 marks]

Question 2c

Predict the shape around each of the C and N atoms in $HCONH_2$.

[2 marks]

Question 2d

State, with a reason, whether $HCONH_2$ is a polar molecule.

[3 marks]

Question 3a

The Pauling electronegativity values of different elements are shown in Fig 3.1.

H 2.1																
Li 1.0	Be 1.6											B 2.0	C 2.5	N 3.0	O 3.5	F 4.0
Na 0.9	Mg 1.3											Al 1.5	Si 1.9	P 2.2	S 2.6	Cl 3.0
K 0.8	Ca 1.0	Sc 1.4	Ti 1.5	V 1.6	Cr 1.7	Mn 1.5	Fe 1.8	Co 1.9	Ni 1.9	Cu 1.9	Zn 1.6	Ga 1.8	Ge 2.0	As 2.2	Se 2.6	Br 2.6

Fig. 3.1

A compound formed from magnesium and oxygen has a different structure to a compound formed from phosphorus and oxygen.

Predict the type of bonds that will occur in each compound. Explain your answer.

[4 marks]

Question 3b

Explain why the melting point of phosphorus(III) oxide, P_4O_6, is lower than that of magnesium oxide in terms of their bonding and structure.

[2 marks]

Question 3c

Phosphorus(III) oxide, P_4O_6, contains no P–P or O–O bonds.

In the P_4O_6 molecule, all oxygen atoms are bonded to two other atoms and all phosphorus atoms are bonded to three other atoms.

Sketch a structure for P_4O_6.

[1 mark]

Question 3d

Explain the difference in electronegativity shown in Fig. 3.1 between magnesium and calcium.

[3 marks]

Question 4a

Phosphorus reacts with chlorine to form PCl_5.

i)
State the shape of a molecule of PCl_5 and give two different bond angles within a molecule of PCl_5.

shape of PCl_5 ....

bond angles in PCl_5 ....

[2]

ii)
Draw the dot-and-cross diagram of a molecule of PCl_5. Show the outer electrons only.

[2]

[4 marks]

Question 4b

PCl_5 is an example of a compound that exists as two structures depending on the conditions.

$$2PCl_5(g) \rightleftharpoons [PCl_4]^+[PCl_6]^-(s)$$

Draw diagrams to suggest the shapes of $[PCl_4]^+$ and $[PCl_6]^-$.

$[PCl_4]^+$	$[PCl_6]^-$

[2 marks]

Question 4c

A student suggests that the PCl_5 molecule has sp^3 hybridisation. Explain why the student is not correct.

[2 marks]

Question 4d

Antimony(V) chloride is another Group 15 chloride. Under standard conditions, it is a liquid made from $SbCl_5$ molecules.

Explain why phosphorus(V) chloride in its solid form has a higher melting point than antimony(V) chloride.

[2 marks]

Question 4e

Under the right conditions, molecules of $SbCl_5$ join together to form Sb_2Cl_{10}.

Complete Fig. 4.1 to show the bonding in Sb_2Cl_{10}.

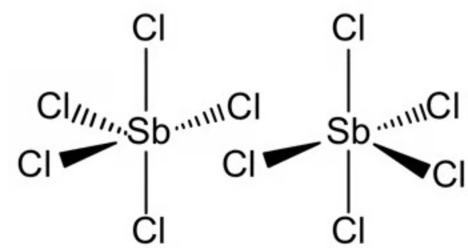


Fig. 4.1

[1 mark]

Question 4f

The other Group 15 elements also form pentachlorides with the exception of nitrogen.

Suggest a reason why nitrogen does not form NCl_5.

[1 mark]

Question 5a

Paperclips have a higher density than water. However, if a paperclip is lowered carefully onto the surface of water, the paperclip can float. When a drop of liquid soap is added to the water, the paperclip sinks.

Suggest explanations for these observations.

[3 marks]

Question 5b

Ammonia is highly soluble in water. Draw diagrams to show the two ways that intermolecular forces can form between a molecule of water and a molecule of ammonia to explain this.

[4 marks]

Question 5c

NH_3, HCl and F_2 all exist as gases at room temperature and pressure.

NH_3 has the highest boiling point of -33 °C .

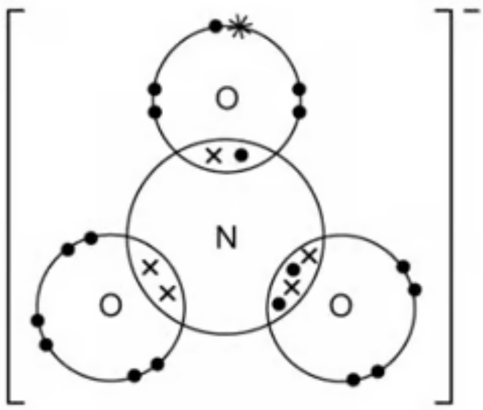
Predict whether HCl or F_2 has the lowest boiling point and explain why there is a difference between their boiling points.

[5 marks]

Model Answers: Hard

1a

a) The dot-and-cross diagram for the NO_3^- ion is:



- Dot and cross correct around N atom showing double, single and a coordinate bond; [1 mark]
- Correct number of electrons for each atom
  AND
  Extra electron for O^-; [1 mark]

[Total: 2 marks]

- **Careful:** Read the question, as it explains the three types of covalent bonds that are in a NO_3^- ion
- As it is a polyatomic ion you need to include [ ] and the negative charge on the outside
- Use your displayed formula to help guide your dot and cross diagram:
- Start by drawing your N atom as normal with 5 electrons in the outer shell
- Then take each type of covalent bond at a time and draw either the single, double or dative covalent bond with the correct number of electrons
  - The single bond will have one electron from the N atom and one from an O atom
    - This O atom needs an extra electron to complete the shell and this is the electron that causes the ion to have a 1– charge

  - The coordinate bond will have two electrons from the N atom
  - The double bond will have two electrons from the N atom and two electrons from an O atom

1b

b) The formulae of the ions present in $H_4N_2O_3$ are:

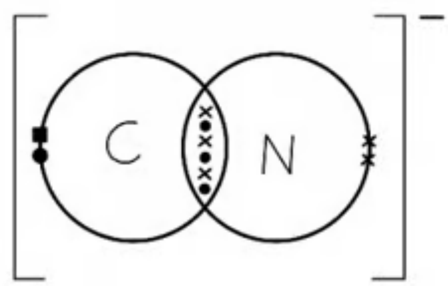
- NH_4^+; [1 mark]
- NO_3^-; [1 mark]

[Total: 2 marks]

- Empirical formula - simplest whole-number ratio of atoms of elements in a compound
- You need to learn all of your polyatomic ion formulae and know them from memory, as they won't be given in the exam
- In this instance, the empirical formula is also the molecular formula and NH_4^+ and NO_3^- make up $H_4N_2O_3$
- Don't forget the correct charges on the ions in your answer as well

1c

c) The dot-and-cross structure for a CN^- ion is:



- Correct bonding shown; [1 mark]
- Rest of the structure correct, including charge; [1 mark]

[Total: 2 marks]

- To deduce the dot-and-cross diagram for the CN^- ion, you need to consider the number of outer electrons of each atom and the overall charge of the ion:

- Carbon has 4 outer electrons
- Nitrogen has 5 outer electrons
- +1 extra outer electron as it is the CN^- ion with a 1– charge

- Carbon and nitrogen must bond together to **form a triple bond** for nitrogen to have a full outer shell of 8 outer electrons
  - As it is a CN^- ion, there must also be an extra electron in the carbon atoms' outer shell for carbon to have a full outer shell of 8 outer electrons
- Don't forget your [ ] and negative charge on the outside of the brackets

1d

d)

i) The average bond enthalpy of the C-N bond in the cyanide ion compared to the C-N bond in the methylamine molecule:

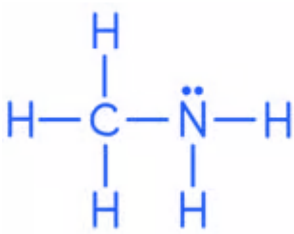
- Average bond enthalpy for the C-N (triple) bond in the cyanide ion is greater / average bond enthalpy of the C-N (single) bond in methylamine is lower; [1 mark]
- Triple bonds are stronger than single and double bonds so require more energy to break; [1 mark]

ii) The C-N bond length in the cyanide ion is shorter than in methylamine because:

- There is a larger negative charge density / electron density between the (carbon and nitrogen) nuclei; [1 mark]
- The greater forces of attraction between electrons and the nuclei pull the nuclei closer together; [1 mark]

[Total: 4 marks]

- To do this question you need to know, or work out, the structure of methylamine first and establish that the C-N bond is a single bond

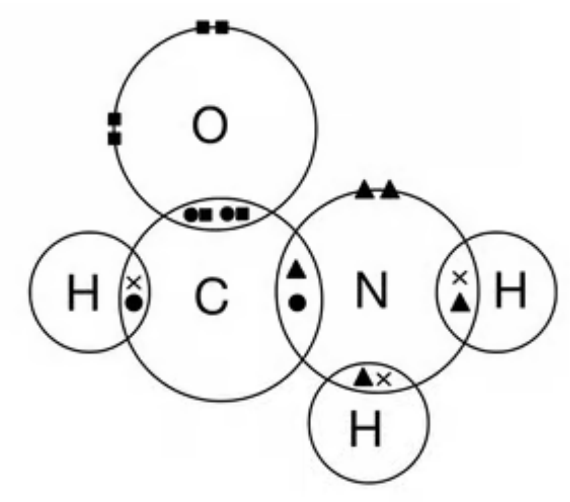


- Where the covalent bond is between the same two atoms, in this case carbon and

nitrogen, the single bond is always the weakest and the triple bond is the strongest
- The bond length is the internuclear distance between two covalently bonded atoms
- The C-N triple bond in the cyanide ion has 6 electrons between the carbon and nitrogen atom, compared with 2 electrons in the C-N single bond in methylamine
- There is a much greater attraction between the negative electrons and the positive nuclei in the cyanide ion, which pulls the atoms closer, decreasing the bond length

2a

a) A dot-and-cross diagram of $HCONH_2$:



- Correct bonding pairs; [1 mark]
- Correct non-bonding pairs; [1 mark]

[Total: 2 marks]

- Don't be put off this question with an unfamiliar compound, the way that you go about drawing the dot-and-cross diagrams is the same
- It is a good idea to learn the number of bonds that common atoms form in covalent compounds when they are not charged:
  - Carbon forms 4 bonds
  - Nitrogen forms 3 bonds
  - Oxygen forms two bonds
  - Hydrogen and halogens form one bond
- Once you have drawn sufficient electrons in each area of overlap, add non-bonding electrons to each atom to ensure that each atom has the correct number of electrons

2b

b) The bond angles in $HCONH_2$ are:

H-C-N:

- Any angle between 117° and 120°; [1 mark]
- Due to three bonded regions / 2 single bonds and 1 double bond (and no lone pairs); [1 mark]
- That repel each other equally; [1 mark]

C-N-H:

- Any angle between 104° and 108°; [1 mark]
- Due to 3 electron / bonded pairs and 1 lone pair; [1 mark]
- Greater repulsion between the unbonded pair / lone pair (on nitrogen) than between unbonded pairs; [1 mark]

[Total: 6 marks]

- **Remember:** Only the atoms that are surrounded by other atoms are 'central' as they need to have at least two different atoms attached in order to have a bond angle

- That repel each other equally; [1 mark]

C-N-H:

- Any angle between 104° and 108°; [1 mark]
- Due to 3 electron / bonded pairs and 1 lone pair; [1 mark]
- Greater repulsion between the unbonded pair / lone pair (on nitrogen) than between unbonded pairs; [1 mark]

[Total: 6 marks]

- **Remember:** Only the atoms that are surrounded by other atoms are 'central' as they need to have at least two different atoms attached in order to have a bond angle
- The bond angle is based on the number of regions of electron density, so includes lone pairs, not just the number of bonded atoms
- Bond angles are reduced by the presence of unbonded electron pairs
- The bond angle that results between four bonded pairs of electrons is 109.5°
- The C-N-H bond angle is predicted to be 107° as the lone pair present pushes the bonded pairs of electrons closer, reducing the bond angle by 2.5°
- In VSEPR, the repulsion between a multiple bond and a single bond is assumed to be the same as the repulsion that occurs between single bonds
- This would give the H-C-N bond angle to be 120°
- In reality, the multiple bond will provide extra repulsion and reduce this bond angle, hence any angle between 117° and 120° is accepted

2c

c) The shapes around the C and N in $HCONH_2$ are:

Around the carbon:

- Trigonal planar; [1 mark]

Around the nitrogen:

- Pyramidal; [1 mark]

[Total: 2 marks]

- The shape around each atom is determined by the number of regions of electron

density around the atom and it is the basis for working out bond angles
- Different bond angles give rise to different shapes, common bond angles and shapes that you need to know are:
  - Trigonal planar: 120°
  - Linear: 180°
  - Tetrahedral: 109.5°
  - Pyramidal, 107°
  - Non-linear: 104.5°
  - Octahedral: 90°
  - Trigonal bipyramidal: 90° and 120°

2d

d) Stating whether $HCONH_2$ is a polar molecule:

- The molecule is polar; [1 mark]
- As there are polar bonds present/ C=O is polar/ N-H is polar; [1 mark]
- There is an unsymmetrical distribution of charge; [1 mark]

[Total: 3 marks]

- For a molecule to be polar:
  - It must have polar bonds
  - These bonds must distribute electron density unsymmetrically across the molecule
- You should know that C=O and N-H are polar bonds arising from the difference in electronegativities between the two atoms involved in the bond

3a

a) The type of bonds that will occur in each compound are:

- Between Mg and O: Ionic bonds; [1 mark]
- Because the difference in electronegativity is high / greater than 2.0 / 2.2; [1 mark]
- Between P and O: Polar covalent bonds
  OR
  Covalent with some ionic character; [1 mark]
- Because the difference in electronegativity between 1.0 and 2.0 / 1.2; [1 mark]

[Total: 4 marks]

- This may seem like a straightforward question but it is an easy place to slip up and lose marks if your reasoning is not good enough
- You need to be aware that the type of bonding and hence the compound formed depends on electronegativity differences between the atoms
  - Generally, a difference in electronegativity of greater than 2.0, the bond is likely to be ionic
  - Between 1.0 and 2.0, the bond is likely to be a polar covalent bond
  - Lower than 1.0, the bond is likely to be covalent
  - A value of 0.0 means there is no difference and the bond will be non-polar
- It is not enough to say "magnesium is a metal and oxygen is a non-metal" and "carbon and oxygen are both non-metals"
  - These types of responses are too basic at this level
  - The presence of the Pauling electronegativity values should indicate that you need to use these in your response

3b

b) The melting point of phosphorus(V) oxide is lower than that of magnesium oxide as:

- MgO has a giant ionic lattice with strong (ionic) bonds
  AND
  P_4O_6 has a simple molecular / covalent structure with weak intermolecular forces; [1 mark]

- More energy is required to break the bonds in MgO than overcome the weak intermolecular forces in P_4O_6; [1 mark]

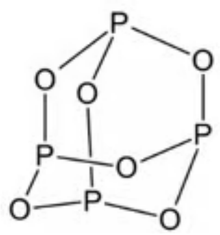
[Total: 2 marks]

- It is very important to make the distinction between bonding within molecules and bonding between molecules
- P_4O_6 has strong intramolecular covalent bonds (within molecules) **but weak** intermolecular forces (between molecules)
- A common mistake is to forget about the intermolecular forces and say that covalent bonds must be weaker than ionic bonds, which is not the case
- You could also use the following terms instead of intermolecular forces to gain this mark:

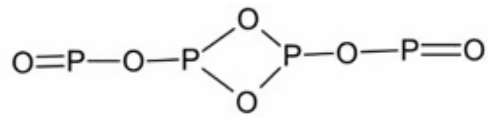
- London dispersion forces
- Van der Waals' forces
- Permanent dipole-permanent dipole (pd-pd) forces

3c

c) A structure for P_4O_6 is:



OR



- P_4O_6 structure where each P has three P-O bonds and each O has two P-O bonds; [1 mark]

[Total: 1 mark]

- The first diagram shows the correct structure, but the second is acceptable as two O atoms have a P=O double bond so is divalent

3d

d) The difference in electronegativity shown in Fig. 3.1 between magnesium and calcium is because:

- Calcium has more shielding (than magnesium); [1 mark]
- Calcium has a larger atomic radius (than magnesium); [1 mark]
- (Therefore) there is less attraction between the nucleus and the bonding pair of electrons (so the electronegativity of calcium is less than magnesium); [1 mark]

[Total: 3 marks]

- Whilst Ca has a greater nuclear charge which would attract the bonding pair of electrons more, this attraction is negligible compared to the decreases in attraction due to the greater shielding and larger atomic radius
- When explaining differences in electronegativity, always think about:
  - Nuclear charge (an important factor in the differences between elements within the same period)

- Atomic radius (an important factor in the differences between elements within the same group)
- Shielding (an important factor in the differences between elements within the same group)

- Then use these factors to explain what effect these have on the attraction between the nucleus and the bonding pair of electrons
- For all of these marking points, you could say the reverse for the mark, e.g. magnesium has less shielding than calcium

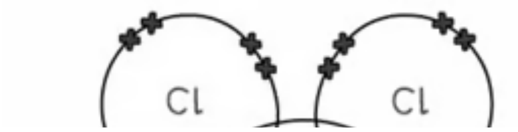
4a

a)

i) The shape and bond angles in a molecule of PCl_5 are:

- Shape: Trigonal bipyramidal; [1 mark]
- Bond angles: 120° **AND** 90°; [1 mark]

ii) The dot-and-cross diagram of a molecule of PCl_5 is:



attraction due to the greater shielding and larger atomic radius

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    - Nuclear charge (an important factor in the differences between elements within the same period)
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    - Shielding (an important factor in the differences between elements within the same group)
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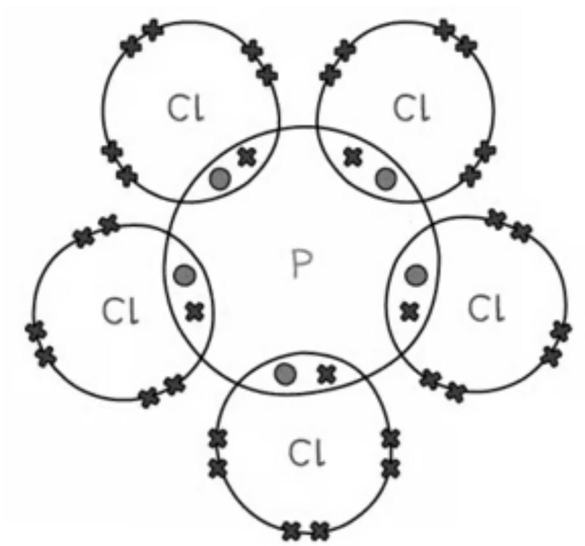


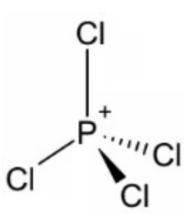
- 5 bonded pairs of electrons; [1 mark]
- Rest of structure correct; [1 mark]

[Total: 4 marks]

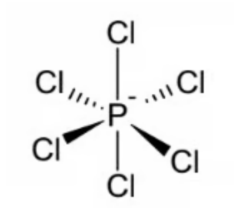
- Phosphorus(V) chloride is an example of when the central atom does not have a noble gas configuration and has expanded its octet of electrons
- Phosphorus has 5 outer electrons and each electron will pair with an electron from a chlorine atom to form 5 covalent bonds, with phosphorus having 10 electrons in its outer shell

4b

b) The diagrams to suggest the shapes of $[PCl_4]^+$ and $[PCl_6]^-$ are:



$[PCl_4]^+$



$[PCl_6]^-$

- Each correct structure; [1 mark]

[Total: 2 marks]

- In $[PCl_4]^+$, the central P atom has 4 bonded pairs of electrons which will repel equally, giving a tetrahedral shape
- In $[PCl_6]^-$, the central P atom has 6 bonded pairs of electrons which repel equally, giving an octahedral shape
    - You can also draw the structures with square brackets around the ion and the charge on the ion
- Make sure that you use wedges and dotted lines to show the three-dimensional shape of each structure

4c

c) PCl_5 does not have sp3 hybridisation because:

- sp3 orbitals are formed from one s orbital and 3 p orbitals; [1 mark]
- But PCl_5 has 5 covalent / sigma / σ bonds; [1 mark]

[Total: 2 marks]

- PCl_5 has 5 covalent bonds formed from the end-on overlap of atomic orbitals forming 5 σ bonds
- Each σ bond will be a hybrid of 5 atomic orbitals
- sp^3 hybridisation occurs when an s orbital and 3 p orbitals overlap, i.e. when 4 orbitals overlap, so PCl_5 cannot have sp^3 hybridisation
- PCl_5 actually has sp^3d hybridisation, but you are not expected to know this and this type of hybridisation is beyond the scope of the course

4d

d) Phosphorus(V) chloride in its solid form has a higher melting point than antimony(V) chloride because:

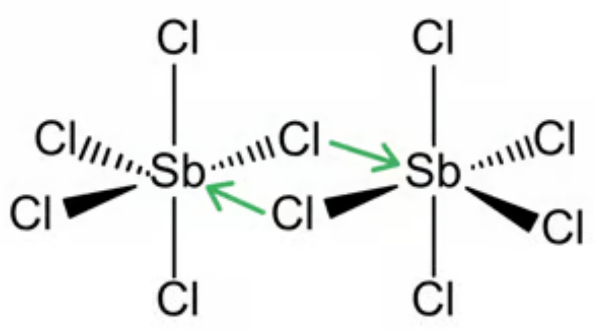
- It / PCl_5 has strong(er) electrostatic forces of attraction between the ions; [1 mark]
- (Than) weak intermolecular forces (in $SbCl_5$); [1 mark]

[Total: 2 marks]

- To explain differences in melting point, you need to think about the structure and bonding within each compound
- Using the equation given in part (b), you can see that when solid, phosphorus(V) chloride exists as two ions, PCl_4^+ and PCl_6^-
- This is key to understanding the type of structure - this means that it has a giant ionic lattice made from the positive and negative ions which have strong ionic bonds between them
- You would not be awarded the first mark for any reference to PCl_5 molecules or intermolecular forces or breaking covalent bonds
- You are told that in antimony(V) chloride exists as a liquid made from $SbCl_5$ molecules, therefore when explaining its melting point, you need to refer to weak intermolecular forces
- You could also use the following terms instead of intermolecular forces to gain this mark:
  - London dispersion forces
  - Van der Waals' forces
  - Permanent dipole-permanent dipole (pd-pd) forces
  - Instantaneous dipole - induced dipole (id-id) forces
- Any reference to the breaking of covalent or ionic bonds is ignored

4e

e) The bonding in Sb_2Cl_{10} is:



- Two coordinate bonds shown as arrows; [1 mark]

[Total: 1 mark]

- You should recognise that $SbCl_5$ has a similar structure to PCl_5 and so is able to expand its octet further, like in the PCl_6^- ion
- A lone pair of electrons from a Cl atom can be donated to an Sb atom on a neighbouring molecule which forms a coordinate bond
- Coordinate bonds are shown by using an arrow, with the arrowhead pointing from the atom where the lone pair of electrons are coming from, i.e. from the Cl atom
- Make sure you know how to represent a coordinate bond; examiners comment that many students are not familiar with this convention

4f

f) Nitrogen does not form NCl_5 because:

Any **one** of the following:

- Nitrogen cannot expand its octet; [1 mark]
- Nitrogen has no (2)d orbitals; [1 mark]
- Nitrogen is too small (to bond to 5 atoms); [1 mark]
- The repulsion between electron pairs would be too great; [1 mark]

[Total: 1 mark]

- This type of question is typically not well answered

- Students seem to lack an understanding that Period 2 elements are unable to contain more than eight electrons in their outer shell because their atoms are too small and they do not have any 2d orbitals for the electrons to occupy - these are worth learning!

5a

a) An explanation for the observations is:

- Water has high surface tension; [1 mark]
- (The paperclip floats because) there is hydrogen bonding between the water molecules; [1 mark]
- (The paperclip sinks because) the soap breaks the hydrogen bonding; [1 mark]

[Total: 3 marks]

- Whenever you need to explain the properties of water, you should think about hydrogen bonding
- The anomalous properties of water are due to the hydrogen bonds that form between the lone pair of electrons of an oxygen atom on one water molecule and a hydrogen atom with a partial positive charge on another water molecule
- You should be aware that the high surface tension of water arises due to hydrogen bonds

that many students are not familiar with this convention

4f

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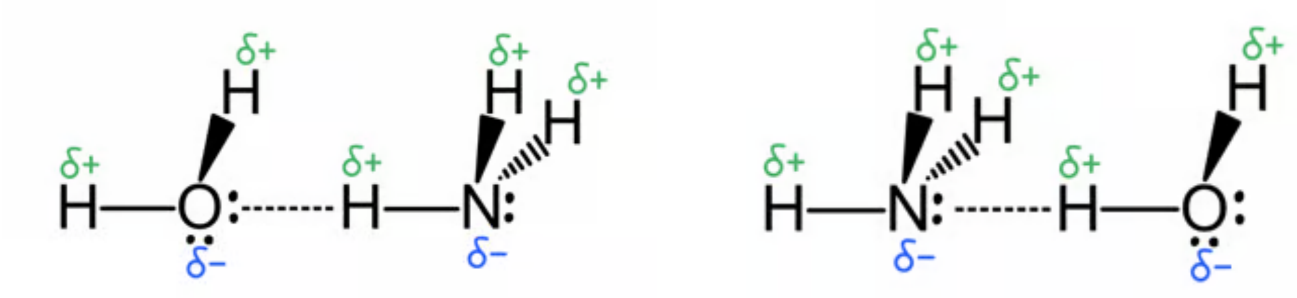
- The anomalous properties of water are due to the hydrogen bonds that form between the lone pair of electrons of an oxygen atom on one water molecule and a hydrogen atom with a partial positive charge on another water molecule
- You should be aware that the high surface tension of water arises due to hydrogen bonds
    - This causes the paperclip to float
- When it sinks, these hydrogen bonds must have been overcome, so the drop of

soap must have caused this to happen
    - You are not expected to know the reason why this occurs
- You must explain both observations to gain full marks, i.e. why it floats initially and then why it sinks when the drop of soap is added

5b

b) Diagrams to show the two ways that intermolecular forces can form between a molecule of water and a molecule of ammonia to explain why ammonia is highly soluble in water are:



- δ- on O atom and at least one δ+ H shown correctly in an atom of water; [1 mark]
- δ- on N atom and at least one δ+ H shown correctly in an atom of ammonia; [1 mark]
- One diagram showing a hydrogen bond between H in an ammonia molecule and a O atom of a water molecule; [1 mark]
- One diagram showing a hydrogen bond between H in a water molecule and a N atom of an ammonia molecule; [1 mark]

[Total: 4 marks]

- You should recognise that ammonia will form hydrogen bonds with water as it contains a hydrogen atom attached to the electronegative nitrogen atom
- Hydrogen bonds can exist between the δ- on N and the δ+ H of a water molecule OR between the δ- on O and the δ+ H of an ammonia molecule
- A hydrogen bond is represented by a dashed line

5c

c) Predict which has the lowest boiling point and explain why there is a difference between their boiling points:

- F_2 has the lowest boiling point; [1 mark]

- F_2 has London dispersion forces; [1 mark]
- HCl also has permanent dipole - permanent dipole / (pd-pd) forces; [1 mark]
- The intermolecular forces are stronger in HCl
  OR
  Permanent dipole - permanent dipole / (pd-pd) forces are stronger than London dispersion forces; [1 mark]
- That require more energy to overcome in HCl
  OR
  Less energy to overcome in F_2; [1 mark]

[Total: 5 marks]

- You need to identify the intermolecular forces that exist within each substance
- Both will have London dispersion forces
- HCl has a polar bond that has a permanent dipole, so will experience permanent dipole-permanent dipole forces
- You must also state the relative strength of each type of intermolecular force and use this to explain which will require the most / least energy to overcome
- You must make sure that you make comparative statements to gain the marks